

Scientific Research Report

Interpretable Ensemble Learning Predicts Antibiotic Resistance in *Treponema denticola* Using Expert ClassifiersPradeep Kumar Yadalam^a, Prabhu Manickam Natarajan^{b*}, Carlos M. Ardila^{a,c,d*}^a Department of Periodontics, Saveetha Dental College and Hospital, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, Tamil Nadu, India^b Department of Clinical Sciences, Center of Medical and Bio-allied Health Sciences and Research, College of Dentistry, Ajman University, Ajman, United Arab Emirates^c Universidad de Antioquia U de A, Medellín, Colombia^d Biomedical Stomatology Research Group, Universidad de Antioquia U de A, Medellín, Colombia

ARTICLE INFO

Article history:

Received 14 March 2025

Received in revised form

4 June 2025

Accepted 8 June 2025

Available online 11 July 2025

Key words:

Periodontitis

Machine learning

Antimicrobial resistance

Treponema denticola

Microorganisms

ABSTRACT

Introduction and Objectives: Antibiotic resistance is a global health concern, contributing to prolonged hospital stays, increased medical costs, and higher mortality rates. Addressing antimicrobial resistance (AMR) in periodontal infections requires targeted therapies and a multifaceted approach. This study aims to predict and classify AMR genomic sequences in *Treponema denticola*, a key pathogen in periodontal disease, using machine learning (ML).

Methods: UniProt FASTA sequences were used to investigate AMR in *T. denticola*. Data were retrieved and preprocessed using the BioPython library in a Jupyter Notebook. A structured approach included data exploration, feature extraction, and visualization. Four classification models – Random Forest, Support Vector Machine (SVM), Gradient Boosting, and Neural Network (Multilayer Perceptron Classifier) – were optimized using specific hyperparameters. Model performance was evaluated using fivefold stratified cross-validation. A Voting Classifier, combining multiple models, was implemented to enhance predictive accuracy.

Results: The Voting Classifier outperformed Random Forest, SVM, Gradient Boosting, and Neural Network models, achieving the highest test accuracy (96.46%) and F1-score (0.9646). High accuracy was also demonstrated by SVM and Neural Networks (95.58%), but the robustness of the Voting Classifier was highlighted by its ability to balance accuracy with low log loss (0.1504). **Conclusion:** This study highlights the effectiveness of the Voting Classifier in classifying AMR genomic sequences in *T. denticola*. The findings underscore the potential of interpretable ML approaches for advancing AMR research in periodontal pathogens and informing targeted therapeutic strategies.

Clinical Relevance: The ability to accurately predict AMR in *T. denticola* using ML models like the Voting Classifier can significantly enhance clinical decision-making. By identifying resistance patterns, clinicians can tailor antibiotic therapies more effectively, reducing treatment failures and mitigating the spread of resistance. This approach also supports the development of novel antimicrobial agents and strengthens public health surveillance efforts, particularly in resource-limited settings where periodontal infections are prevalent.

© 2025 The Authors. Published by Elsevier Inc. on behalf of FDI World Dental Federation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

* Corresponding author. Department of Clinical Sciences, Center of Medical and Bio-allied Health Sciences and Research, College of Dentistry, Ajman University, Ajman 346, United Arab Emirates; and Department of Basic Sciences, Faculty of Dentistry, Universidad de Antioquia, Calle 70 No. 52-21, Medellín 050010, Colombia.

E-mail addresses: prabhuperio@gmail.com (P.M. Natarajan), martin.ardila@udea.edu.co (C.M. Ardila).

Pradeep Kumar Yadalam: <http://orcid.org/0000-0002-6653-4123>

Carlos M. Ardila: <http://orcid.org/0000-0002-3663-1416>

<https://doi.org/10.1016/j.identj.2025.100884>

0020-6539/© 2025 The Authors. Published by Elsevier Inc. on behalf of FDI World Dental Federation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Introduction

Antibiotic resistance is a significant global health threat, causing extended hospital stays, higher medical costs, and increased mortality.¹ The 2019 Antibiotic Resistance Threats Report revealed that over 2.8 million antibiotic-resistant infections occur in the United States, resulting in over 35,000 deaths per year.² Government agencies, such as the US Food and Drug Administration, actively endorse the One Health Initiative, which advocates for a comprehensive approach to combating antibiotic resistance. Periodontal infections, encompassing a spectrum of inflammatory conditions that affect the supporting structures of the teeth, constitute a significant public health concern worldwide.³ These diseases, primarily characterized by gingivitis and periodontitis, are precipitated by the accumulation of dental plaque and are influenced by a complex interplay of microbial, host, and environmental factors.⁴ The World Health Organization estimates that severe periodontitis affects approximately 10% to 15% of the global population, highlighting its prevalence as a major contributor to oral morbidity.⁵

Periodontal infections are prevalent in developing countries due to socioeconomic and healthcare access barriers. Limited awareness of oral health, inadequate dental care infrastructure, and insufficient public health initiatives exacerbate the situation. Cultural attitudes towards oral health often led to neglect and underreporting of oral diseases. A study found that the prevalence of periodontal diseases in developing regions can exceed 70% among adults.⁶ Healthcare systems face challenges, including a shortage of trained dental professionals and limited access to preventive care, which leads individuals to prioritize urgent health needs over routine dental visits. Disparities in socioeconomic status also affect access to necessary oral hygiene products, impairing the ability to maintain adequate oral health.⁷ Antimicrobial resistance (AMR) is a concerning issue in developing countries, where antibiotics can be procured without prescriptions, leading to misuse and overuse. Bacteria associated with periodontal infections, such as *Porphyromonas gingivalis* and *Aggregatibacter actinomycetemcomitans*, exhibit increasing resistance to commonly used antibiotics, making standard treatment regimens less effective. Addressing the underlying socioeconomic determinants of health is crucial in reducing the burden of periodontal disease.

Treponema denticola is a gram-negative, anaerobic spirochete bacterium and a key member of the red complex, a group of bacteria associated with the destruction of periodontal tissue. AMR, caused by genetic mutations, gene transfer, and biofilm formation, is a growing concern affecting conventional periodontal therapy, potentially leading to treatment failures.^{4,8} This resistance complicates treatment regimens, often requiring alternative therapeutic strategies. Understanding the resistance mechanisms and genomic sequences of these bacteria is crucial for developing targeted therapies to manage periodontal infections effectively. A multifaceted approach is necessary to address AMR in *T. denticola* infections, encompassing antimicrobial stewardship, combination therapy, adjunctive therapies, and ongoing research and surveillance. Continued research and innovative treatment strategies are crucial for combating the impact of AMR in *T.*

denticola and enhancing the management of periodontal infections.^{9,10}

AMR in bacteria such as *T. denticola* is a growing concern in clinical settings, particularly in periodontal diseases. Predicting AMR gene sequences using machine learning (ML) is crucial for public health protection, surveillance, targeted interventions, drug development, and understanding the mechanisms of resistance. ML models can rapidly classify gene sequences, providing valuable insights for clinical decision-making and enhancing public health surveillance. Advanced methods for early detection and classification of resistant genes are required to address AMR in *T. denticola*. Traditional alignment-based approaches, which frequently fall short in identifying species-specific resistance genes, can be replaced by scalable and data-driven ML models. ML frameworks can improve the prediction of resistance phenotypes by utilizing amino acid composition and other sequence-derived features.

Despite numerous studies investigating ML in AMR prediction, many models are constrained by poor generalizability across datasets, organism-specific training, and a lack of interpretability. Few have focused on *T. denticola* in particular, and none have applied a Mixture of Experts (MoE) strategy to combine the advantages of multiple algorithms for reliable classification. They are trained and validated using unseen data, allowing for real-time predictions for protein sequences.¹¹

Antibiotic resistance genes (ARGs) are crucial for monitoring antibiotic resistance. The two primary methods are sequence-based assembly and alignment, as well as ML techniques. However, these methods have a high false-negative rate for species-specific AMR genes. Proteins can be represented as protein languages inspired by natural language processing to uncover novel arginine-rich regions. A universal ARG prediction framework is needed that incorporates multiomics data from various genomics technologies. Protein language model (PLM)-ARG¹² was introduced to discriminate ARGs based on sequence similarity. A study highlights the global challenge of antibiotic resistance to public health and the environment. While efforts have focused on identifying ARGs, recent research using metagenomic and metatranscriptomic approaches has revealed a notable issue: many proteins, including ARGs, remain unannotated due to sequence differences. A new artificial intelligence-powered framework, PLM-ARG, was proposed to identify ARGs and classify resistance categories using a pretrained large PLM, addressing this issue. The model utilizes extensive data (over 28,000 ARGs and 29 resistance categories) and demonstrates Matthew's correlation coefficient of 0.98360 ± 0.001 in validation, with an impressive Matthew's correlation coefficient of 0.838 when tested against an independent set, surpassing existing ARG prediction tools by 51.8% to 107.9%.^{12,13} Additionally, the model's effectiveness was shown by successfully annotating resistance in the UniProt database and assessing the impact of ARGs on environmental microbiota.

A previous study demonstrates the effectiveness of an ML approach, coupled with a game-theory-based feature evaluation algorithm, for identifying AMR genes in Gram-positive bacteria, specifically those encoding resistance to bacitracin and vancomycin. The model achieved classification

accuracies between 87% and 90%, showcasing the potential of this method as an alternative to traditional alignment-based techniques, particularly when dealing with unculturable pathogens.¹³ Furthermore, a standalone software tool implementing the algorithm and model is presented, making this approach accessible for broader use in the identification of AMR genes. To address the increasing global threat of antibiotic resistance, a novel deep learning approach, DeepARG, has been developed to enhance the accuracy of identifying ARGs in environmental samples using next-generation sequencing. This method utilizes two models, DeepARG-SS and DeepARG-LS, for short and full-length sequences, respectively, and incorporates a dissimilarity matrix of all known ARG categories.^{14–16} Evaluation across 30 antibiotic resistance categories demonstrated high precision and recall compared to traditional best-hit methods, resulting in lower false-negative rates. The associated DeepARG-DB database expands current ARG repositories with high-confidence predictions, offering a more comprehensive resource for AMR annotation and identification through command-line and web-based services. These studies have demonstrated the significant potential of employing ML techniques to enhance the accuracy and efficiency of sequence prediction, particularly in the context of *T. denticola* resistance. Predicting and classifying resistant genetic sequences is crucial for effective infection treatment planning and management. An MoE approach provides an innovative method for analysing genomic sequences related to AMR. By leveraging the expertise of multiple specialized models that focus on different aspects of data, MoE can manage the complexity of biological data. These studies have shown how using ML techniques, including Support Vector Machines (SVM), ensemble classifiers (eg, Random Forest [RF], Gradient Boosting), PLM (eg, PLM-ARG), and deep learning models (eg, DeepARG), can significantly improve the precision and effectiveness of sequence-based AMR prediction. These models classify antibiotic-resistant genes using features such as amino acid composition, sequence embeddings, and similarity matrices, particularly in organisms like *T. denticola*, where traditional alignment-based methods may not be effective. This integration helps mitigate overfitting and enhances generalization to unseen data, improving the accuracy of AMR genetic sequence classification.

ML models for predicting antibiotic resistance often rely on general-purpose frameworks trained on diverse bacterial datasets, which may overlook new sequence variations that conflict with established resistance gene families or species-specific resistance mechanisms. Many models lack ensemble learning techniques or are black-box models with limited interpretability. This study fills these gaps by using an MoE ensemble and combining multiple algorithmic viewpoints for the specific classification of antibiotic resistance-related sequences in *T. denticola*. The framework is assessed using stratified cross-validation and integrates explainable outputs through feature importance analysis. This personalized, biologically-based ML solution focuses on a clinically relevant oral pathogen, emphasizing interpretability and model generalization. This approach facilitates a deeper understanding of the underlying biological mechanisms. It may enable the development of targeted strategies for combating resistance in *T. denticola*, a notable pathogen in periodontal disease. By

leveraging advanced algorithms and data analysis, researchers can uncover patterns and trends that were previously difficult to identify, leading to more effective interventions. We aim to predict and classify the AMR genomic sequences in *T. denticola*.

Materials and methods

Data retrieval

Using UniProt,¹⁷ FASTA sequences related to AMR for *T. denticola* were retrieved. Both positive (resistant) and negative (regular) sequences were obtained through searching. A total of 304 resistant (positive) sequences and 261 regular (negative) sequences were identified from the *T. denticola* genome. The following preprocessing steps were then performed: data cleaning, sequence alignment, and feature extraction.

The BioPython library is installed in a Jupyter Notebook, transforming sequence data into a pandas DataFrame. The function reads compressed FASTA files, processes each record, and manages exceptions. AAC was chosen because it captures crucial compositional patterns associated with AMR and strikes a balance between computational efficiency and biological interpretability. Sequences with ambiguous amino acids or fewer than 50 residues were not included. A 90% sequence identity threshold, as determined by CD-HIT, was used to reduce redundancy. Preprocessing involves evaluating sequence length statistics, analysing data variation, and handling biological sequence data using a structured approach. Using keyword-based filtering and annotation parsing, sequences classified as resistant were identified by searching for terms such as ‘antibiotic resistance’, ‘multidrug resistance’, ‘efflux pump’, or specific names of AMR protein families (eg, Tet, Erm, Msr, or Van proteins). The process involves visualizing sequence length distributions using a boxplot, extracting amino acid features, saving them to a CSV file, and training a Label Encoder on the features and labels. The data was split into an 80% training set and a 20% test set and feature scaling was performed using StandardScaler. Based on manual tuning guided by validation performance across several trials, hyperparameters were chosen. For important parameters such as tree depth, learning rate, and hidden layer size, a restricted grid search was conducted. To maintain enough data for evaluating generalization, an 80/20 train-test split was selected. While preserving class balance, fivefold stratified cross-validation produced reliable performance estimates. Using scikit-learn v1.3, BioPython v1.81, NumPy v1.24, and pandas v2.1 in a Jupyter Notebook environment, all analyses were conducted in Python 3.10.

Model architecture

Four optimized classifiers were used in the development of a supervised ML framework to classify antibiotic-resistant vs regular *T. denticola* protein sequences: RF, SVM, Gradient Boosting Classifier (GBC), and Multilayer Perceptron (MLP) Neural Network. For every protein sequence, 20-dimensional feature vectors were created using the amino acid composition. To maintain class distribution and produce accurate

performance metrics, the dataset was divided into an 80% training set and a 20% test set. Stratified fivefold cross-validation was then used. The RF classifier was set up with bootstrap aggregation enabled, 100 decision trees ($n_estimators = 100$), unlimited depth ($max_depth = None$), at least two samples to split ($min_samples_split = 2$), and one sample per leaf ($min_samples_leaf = 1$). To guarantee reproducibility, $random_state = 42$ was used to seed the model. To enable probability output for soft voting, the SVM model employed a radial basis function kernel ($kernel = 'rbf'$) with a regularization parameter of $C = 1.0$, automatic scaling of $gamma = 'scale'$, and $probability = True$. A learning rate of 0.1, a tree depth of 3 ($max_depth = 3$), 100 boosting stages ($n_estimators = 100$), full subsampling ($subsample = 1.0$), and $random_state = 42$ were used to fine-tune the GBC. Two hidden layers of 50 and 25 neurons ($hidden_layer_sizes = (50, 25)$), ReLU activation, the Adam solver, an L2 regularization term ($alpha = 0.0001$), an adaptive learning rate, and a maximum of 1000 training iterations were all included in the configuration of the MLP Neural Network. An MoE ensemble strategy was employed to implement a Voting Classifier, combining the advantages of these separate models. Soft voting was used by the ensemble, in which the final classification was determined by averaging the predicted class probabilities from each base model. To ensure that each model contributed equally, the base estimators – RF, SVM, GBC, and MLP – were not explicitly weighted ($weights = None$). The scaled training set was used to train this ensemble, and the independent test set was used to assess it. The Voting Classifier was extremely well-suited for the intricate and biologically variable features found in AMR genomic data because of its diversity and aggregation capabilities, which reduced variance and improved predictive robustness.

This study evaluated ML models for biological sequence classification, examining test accuracies, receiver operating characteristic (ROC) curves, and area under the curve (AUC) values. The results showed that the Voting Classifier's accuracy was superior. The systematic approach demonstrated the effectiveness of combining experts in complex biological

classification tasks. Ensemble models significantly improved predictive accuracy in biological sequence classification. This study also emphasizes the importance of assessing feature importance in biological contexts, utilizing RF to generate a plot that showcases feature importances. This visualization facilitates understanding of model behaviour and identifies key features for further research.

Results

We have two datasets of *T. denticola* protein sequences:

1. Regular sequences: 261 proteins
2. Resistant sequences: 304 proteins

The dataset contains 565 sequences with an average length of 345.91 units. The standard deviation of 196.99 indicates considerable variability in sequence lengths, with some being much shorter or longer than the average. The shortest sequence is 37 units long, while the 25th percentile indicates that 25% of sequences have a length of 178 units or shorter. The median length is 312 units, with 75% of sequences falling within this range. The longest sequence is 1424 units, indicating that some sequences are significantly longer than others. The high standard deviation confirms the presence of variability in sequence lengths.

Figure 1 illustrates the boxplot highlighting potential differences in sequence length distributions between regular (negative) and resistant (positive) *T. denticola* sequences. Resistant sequences exhibit a broader range of lengths compared to regular sequences. The central line within each box represents the median sequence length, while the box boundaries indicate the interquartile range. Whiskers extend to 1.5 times the interquartile range, and outliers are represented as individual points. This visualization suggests that sequence length may serve as a distinguishing feature between regular and resistant sequences, warranting further investigation into its potential role in AMR.

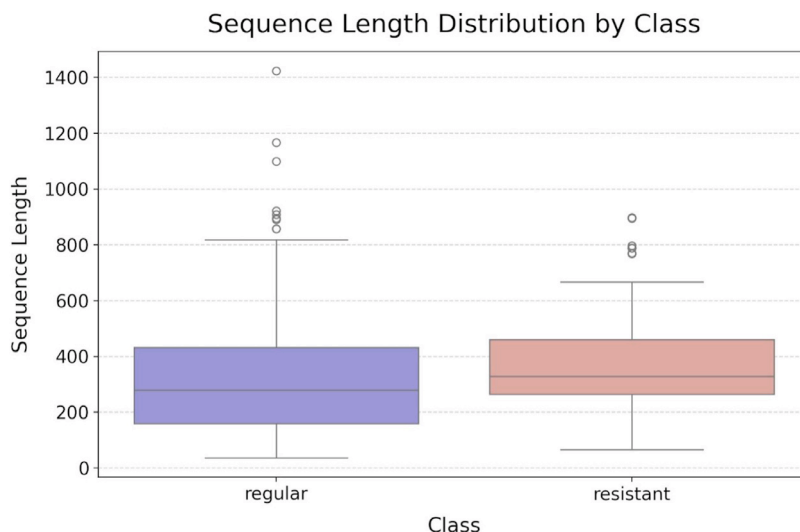


Fig. 1 – Distribution of sequence lengths for regular and resistant *Treponema denticola* sequences.

The amino acid composition of a protein sequence is crucial for understanding its properties and functions. The average proportion of certain amino acids, such as alanine (A), cysteine (C), aspartic acid (D), glutamic acid (E), phenylalanine (F), glycine (G), histidine (H), isoleucine (I), and lysine (K), is approximately 8.18%, 8.19%, 1.19%, 4.55%, 4.55%, 5.16%, and 5.16% respectively. Alanine is a nonpolar, aliphatic amino acid that plays key roles in protein structure and metabolism. Cysteine has a lower average proportion, forming disulfide bonds and stabilizing the three-dimensional structure of proteins. Aspartic acid, a negatively charged amino acid, is involved in enzymatic reactions and functions as a neurotransmitter. Glutamic acid, with a higher average ratio, plays essential roles in cellular metabolism and neurotransmission, particularly in the central nervous system. Phenylalanine is a hydrophobic amino acid precursor to several important biomolecules, including neurotransmitters. Glycine, the smallest amino acid, is key to structural flexibility and is abundant in collagen. Histidine, a polar amino acid, is often found in active sites of enzymes due to its ability to undergo protonation. Isoleucine, a nonpolar and branched-chain amino acid, plays crucial roles in energy production and haemoglobin function. Lysine, a basic amino acid, plays a crucial role in protein synthesis and collagen cross-link formation.

Figure 2 illustrates the bar chart depicting the average frequency of each amino acid in regular (negative) and resistant (positive) *T. denticola* sequences. Notable differences in amino acid frequencies are observed between the two classes, with certain amino acids showing higher or lower prevalence in resistant sequences compared to regular sequences. These variations in amino acid composition likely contribute to the ability of computational models to effectively distinguish between regular and resistant sequences, highlighting their potential role in AMR mechanisms. The chart provides insights into the compositional biases that may underlie functional differences between the two classes.

The study evaluated ML algorithms for predicting and classifying AMR sequences in *T. denticola*. The results showed that the Voting Classifier demonstrated strong performance, with a cross-validation accuracy of $95.13\% \pm 2.18\%$ and a test accuracy of 93.81%. This indicates good stability and reliability across different data splits. The SVM model had a lower cross-validation accuracy of $88.94\% \pm 3.02\%$ but achieved a higher test accuracy of 95.58%. Gradient Boosting achieved a cross-validation accuracy of $92.04\% \pm 2.57\%$, indicating consistent performance across training and testing phases. The Neural Network presented a cross-validation accuracy of $90.50\% \pm 2.74\%$, with the same test accuracy of 95.58% as SVM. However, its higher log loss (0.3579) suggests a greater confidence in misclassification than other models. The Voting Classifier stood out with a test accuracy of 96.46% and no provided cross-validation accuracy, indicating it successfully leverages multiple classifiers and outperforms the others in terms of overall accuracy. The F1 score, a crucial metric for classification tasks, demonstrates the ability to balance recall (sensitivity) and precision. The Voting Classifier emerged as the top performer in predicting and classifying AMR sequences in *T. denticola*, striking a balance between high accuracy and low log loss, while underscoring its robustness. The Voting Classifier, which aggregates predictions from multiple base learners (RF, SVM, Gradient Boosting, and Neural Network), outperformed individual models in terms of accuracy, F1 score, and log loss. Unlike single models, the Voting Classifier benefits from model diversity, effectively balancing bias and variance. For instance, while SVM performed well in high-dimensional spaces, it may struggle with noisy data. RF is robust to overfitting but may underperform on complex, nonlinear patterns. Neural Networks can capture intricate relationships but are prone to overconfidence when insufficient data is available. By integrating their outputs through soft voting, the Voting Classifier leverages the strengths of each model and mitigates individual weaknesses, resulting in

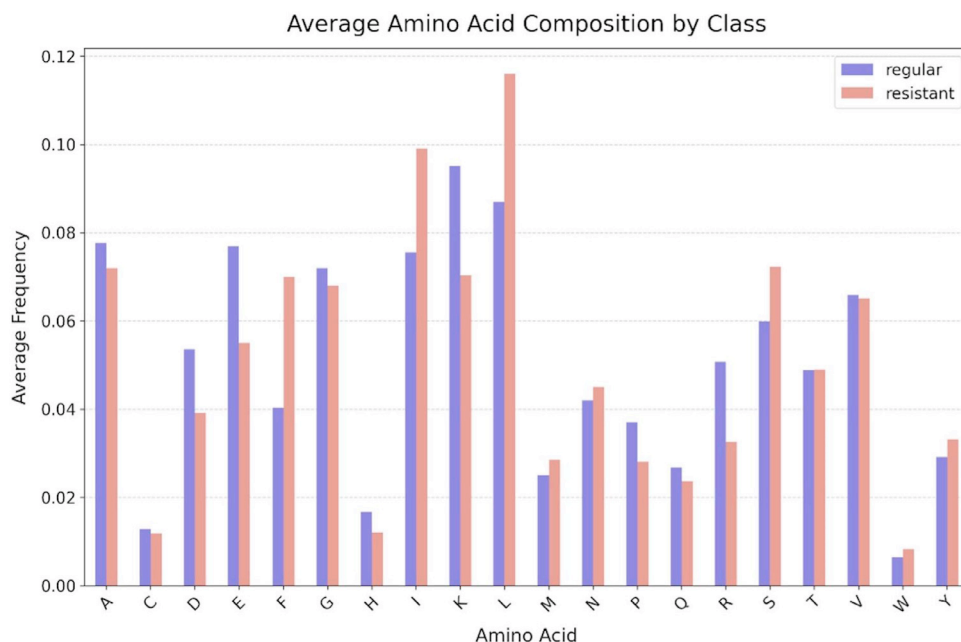


Fig. 2 – Average frequency of amino acids in regular and resistant *Treponema denticola* sequences.

Table – Performance metrics of machine learning models for distinguishing regular and resistant *Treponema denticola* sequences.

Model	Cross-validation accuracy	Test accuracy	F1 score	Recall	Log loss
Random Forest	0.9513 $\pm$ 0.0218	0.9381	0.938140669	0.938053097	0.203224146
SVM	0.8894 $\pm$ 0.0302	0.9558	0.955780074	0.955752212	0.163920221
Gradient Boosting	0.9204 $\pm$ 0.0257	0.9204	0.920466574	0.920353982	0.185176966
Neural Network	0.9050 $\pm$ 0.0274	0.9558	0.955780074	0.955752212	0.357903424
Voting Classifier	N/A	0.9646	0.96460177	0.96460177	0.150406363

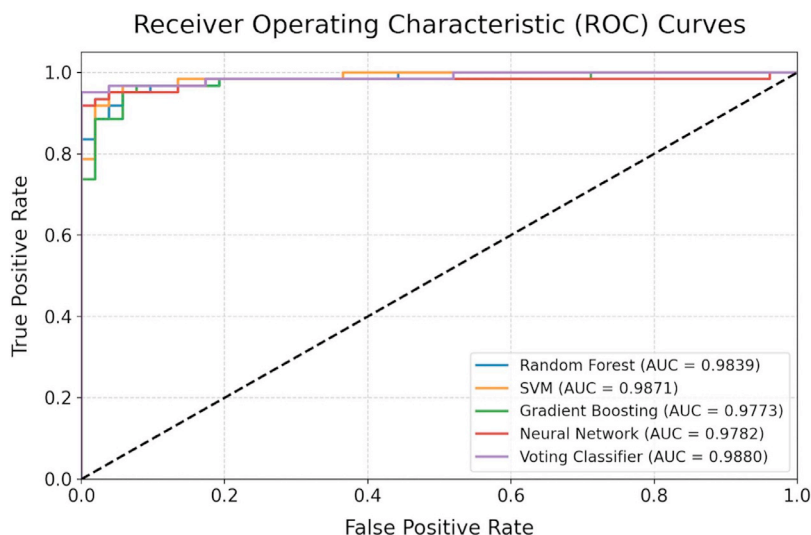
improved predictive performance and generalizability – especially important for complex and heterogeneous biological datasets like AMR sequences. Given the consistently high performance of the individual models, SVM and MLP (both 95.58%), and the high test accuracy of the Voting Classifier (96.46%), it appears that the predictive framework successfully captured discriminative features linked to AMR in *T. denticola*. Model robustness was improved, and the risk of overfitting was decreased by using an ensemble approach that integrated several algorithmic viewpoints. The model's stability across different data splits was further confirmed by stratified cross-validation. There are still certain restrictions, though. The comparatively small dataset size may restrict generalizability, and the diversity of real-world sequences may introduce unexpected variability. While SVM and Neural Network models exhibited high test accuracy, their performance varied based on the data characteristics. RF remained a strong contender, demonstrating stability and reliability for sequence classification and prediction.

Table presents the performance metrics of ML models, including RF, SVM, Gradient Boosting, Neural Network, and a Voting Classifier, for classifying regular and resistant *T. denticola* sequences. Metrics reported include cross-validation accuracy (mean $\pm$ standard deviation), test accuracy, F1 score, recall, and log loss. The Voting Classifier achieved the highest test accuracy (0.9646) and F1 score (0.9646), while the RF model demonstrated strong performance with a cross-validation accuracy of 0.9513 ± 0.0218 . These results highlight

the effectiveness of the MoE approach in distinguishing between the two classes, with the Voting Classifier emerging as the most robust model overall.

Figure 3 presents the ROC curves for the evaluated ML models, including RF, SVM, Gradient Boosting, Neural Network, and the Voting Classifier. The ROC curves illustrate the trade-off between true positive rate (sensitivity) and false positive rate (1-specificity) across different classification thresholds. The Voting Classifier demonstrates superior performance, achieving the highest accuracy and the largest AUC, indicating its effectiveness in distinguishing between regular and resistant *T. denticola* sequences. This visualization underscores the robustness of the Voting Classifier compared to the individual models, highlighting its potential as the optimal choice for this classification task.

Figure 4 shows the confusion matrices for the evaluated ML models, including RF, SVM, Gradient Boosting, Neural Network, and the Voting Classifier, in classifying regular vs resistant *T. denticola* sequences. Each matrix displays the distribution of true positives, true negatives, false positives, and false negatives, providing a detailed assessment of model performance. The Voting Classifier demonstrates the best balance between true positives and true negatives, indicating its superior ability to classify both regular and resistant sequences correctly. This result highlights the Voting Classifier as the most reliable model for this classification task, outperforming the individual models in terms of overall accuracy and robustness.

**Fig. 3 – Receiver operating characteristic (ROC) curves for machine learning models in classifying *Treponema denticola* sequences.**

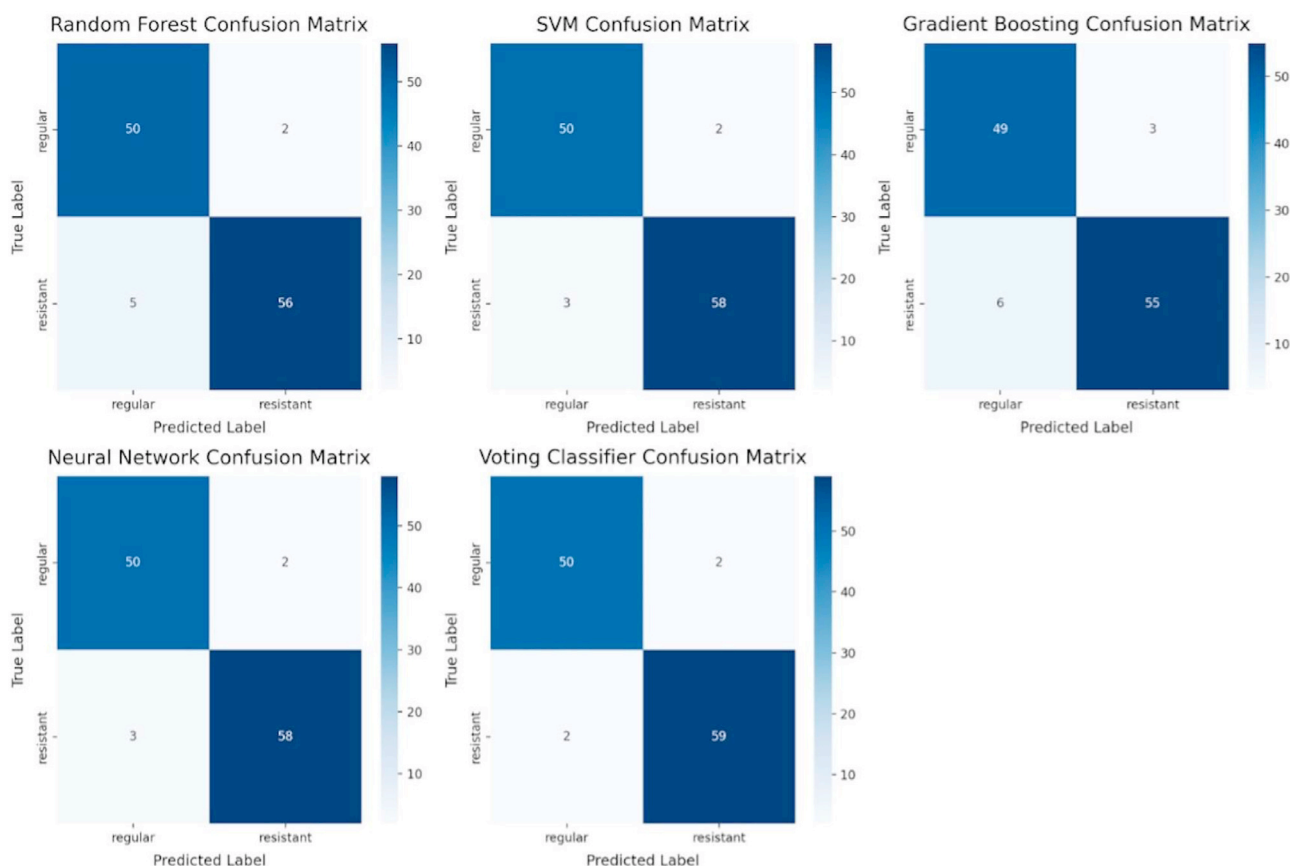


Fig. 4—Confusion matrices for machine learning models in classifying regular and resistant *Treponema denticola* sequences.

Figure 5 depicts the feature importance analysis derived from the RF model, highlighting the amino acids that contribute most significantly to the classification of regular and resistant *T. denticola* sequences. The top amino acids identified as key differentiators are lysine (K), glutamic acid (E), and alanine (A), suggesting their potential role in distinguishing between the two classes. This analysis highlights the significance of specific amino acids in AMR mechanisms and offers insights into the molecular features that underpin the model's decision-making process. The findings align with the hypothesis that compositional differences in amino acids may underlie functional distinctions between regular and resistant sequences.

Discussion

Periodontal disease, a prevalent oral health issue, is primarily caused by the proliferation of bacterial colonies within dental plaque. Antimicrobial therapy is crucial in managing periodontal disease, but its effectiveness depends on understanding the intricate patterns of antimicrobial susceptibility and resistance demonstrated by *T. denticola*. A previous study found that most samples contained more than one ARG for resistance to aminoglycosides, fluoroquinolones, chloramphenicol, rifamycins, and efflux pumps. The most prevalent genes were *msrD*, *mefA*, *cfxA*, and *ermF*, with *mefA*, *msrD*,

cfxA, *ermB*, *tetM*, and *tetQ* for H and C, and *pgpB*, *tetQ*, *cfxA*, *tet32*, *mefA*, and *msrD* for periodontitis samples had a lower prevalence of most ARGs except *pgpB*, which is the most prevalent gene coding for peptide antibiotic resistance. Predicting the sequences of genes responsible for antibiotic resistance in *T. denticola* is crucial and can significantly change the course of drug development.^{4,8,16,18}

This study examines AMR in periodontal pathogens, specifically *T. denticola*, a severe pathogen associated with periodontitis. It uses a Voting Classifier and an expert's approach to identify and classify resistant sequences. Voting Classifier, an ML model that employs an MoE, surpassed individual models in classifying antimicrobial-resistant sequences in *T. denticola*. It achieved a test accuracy of 96.46% by utilizing the strengths of multiple algorithms, including SVM, RF, and Neural Networks. This ensemble approach alleviates the weaknesses of individual models through a soft voting mechanism, making it particularly valuable for complex biological data. SVM and Neural Network achieved test accuracies of 95.58% and 94.69%, respectively (Figures 2-5). However, the higher log loss of the Neural Network indicates potential overconfidence in its predictions. Despite having a lower test accuracy, the RF model demonstrated impressive stability, with a cross-validation accuracy of $92.04\% \pm 2.57\%$, reflecting its robust generalization capabilities. Like previous studies, it presented a novel artificial intelligence-powered language model framework, PLM-ARG,¹¹ for classifying ARG and

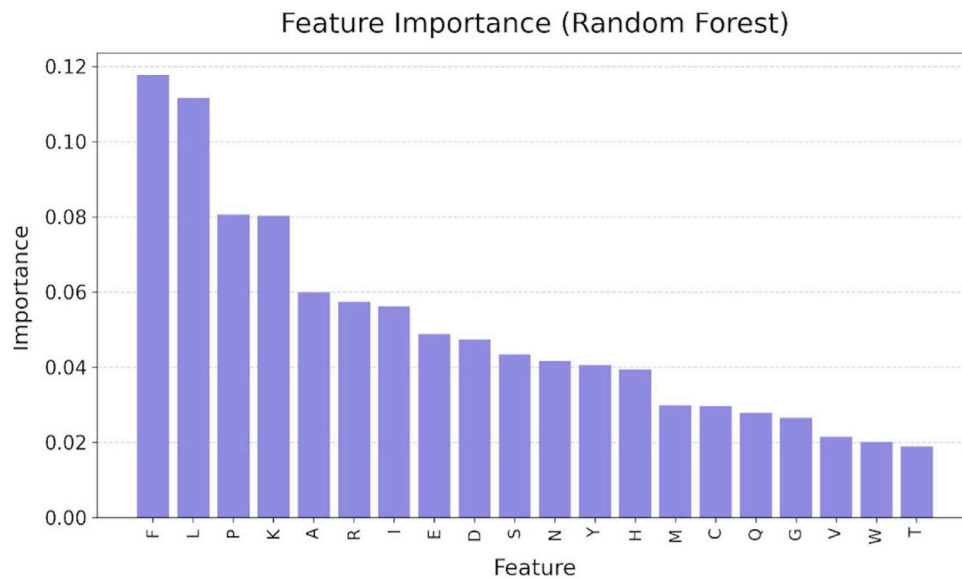


Fig. 5 – Feature importance analysis of amino acids in classifying regular and resistant *Treponema denticola* sequences using the Random Forest model.

resistance categories by translating sequence data into protein language. Based on feature importance analysis using the RF classifier, it was determined that the most crucial amino acids for differentiating resistant from regular *T. denticola* sequences were lysine (K), glutamic acid (E), and alanine (A). These residues play significant structural and functional roles in the context of AMR. Lysine (K), a positively charged amino acid, is known to be abundant in antimicrobial peptides and resistance-granting motifs. It frequently mediates interactions with negatively charged bacterial membranes and affects the function of efflux pumps. Glutamic acid (E), a negatively charged residue, is crucial for protein folding and the distribution of active site charges, especially in resistance-associated enzymes such as ribosomal protection proteins and β -lactamases. Alanine (A), which is small and nonpolar, contributes to conformational flexibility and hydrophobic core packing, both of which can impact the stability of proteins in resistant strains. These findings are consistent with earlier studies that demonstrate how biases in the amino acid composition of AMR-related proteins can lead to structural and functional alterations associated with resistance phenotypes. Therefore, their significance in our model aligns with biological mechanisms that are known to support resistance. The proposed method outperformed state-of-the-art approaches by an improvement range of 9.6% to 36% in AUC and 40.8% to 107.3% in F1-score, as determined by the independent validation set. This study outperformed SOTA-PLM ARG¹²⁻¹⁵ for specific organisms, such as *T. denticola*. Also, it revealed that amino acids lysine, glutamic acid, and alanine are crucial for distinguishing between resistant and nonresistant *T. denticola* sequences. These amino acids influence protein function, cellular metabolism, enzyme active sites, and the structure of peptidoglycan.

Variations in these proportions may suggest evolutionary adaptations related to AMR. A few limitations of this study

include the fact that ML models for AMR in *T. denticola* have several limitations, including data quality and quantity of sequences, advanced feature selection, and the dynamic nature of AMR. However, computational resources for sophisticated models, such as ensemble methods and Neural Networks, will improve the results. Future research should enhance the interpretability of ML models by incorporating multiomics data, handling imbalanced datasets, conducting real-world validation, updating and adapting models, and conducting comparative studies across the oral and periodontal microbiome. Understanding which features contribute to predictions is crucial in clinical settings, and incorporating the fusion of multiomics with metagenomics data can enhance prediction accuracy and provide deeper insights into AMR mechanisms. Real-world validation and continuous learning methods can also be explored to maintain accuracy over time.

Conclusion

The study demonstrates that a Voting Classifier method, which combines multiple algorithmic perspectives, effectively classifies AMR sequences in *T. denticola*, offering insights into potential resistance mechanisms. This represents a step towards interpretable ML in the context of periodontal genomic resistance, and future research should explore more sophisticated explainability techniques.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Author contributions

Prabhu Manickam Natarajan, Pradeep Kumar Yadalam, and Carlos M. Ardila contributed to the conception, analysis, interpretation of data, and drafting of the manuscript. Prabhu Manickam Natarajan, Pradeep Kumar Yadalam, and Carlos M. Ardila: Conceptualization. Prabhu Manickam Natarajan, Pradeep Kumar Yadalam, and Carlos M. Ardila: Methodology. Prabhu Manickam Natarajan, Pradeep Kumar Yadalam, and Carlos M. Ardila: Data curation. Pradeep Kumar Yadalam, and Carlos M. Ardila: Writing – original draft preparation. Prabhu Manickam Natarajan, Pradeep Kumar Yadalam, and Carlos M. Ardila: Visualization. Pradeep Kumar Yadalam, and Carlos M. Ardila: Investigation. Prabhu Manickam Natarajan, Pradeep Kumar Yadalam, and Carlos M. Ardila: Validation. Pradeep Kumar Yadalam, and Carlos M. Ardila: Writing – reviewing and editing.

Acknowledgements

We would like to thank the Center of Medical and Bioallied Health Sciences and Research, Ajman University, Ajman, UAE.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

REFERENCES

1. Ho CS, Wong CTH, Aung TT, et al. Antimicrobial resistance: a concise update. *Lancet Microbe* 2025;6(1):100947.
2. Flynn CE, Guarner J. Emerging antimicrobial resistance. *Mod Pathol* 2023 Sep;36(9):100249. doi: 10.1016/j.modpat.2023.100249.
3. Hu H, Yao Y, Liu F, et al. Integrated microbiome and metabolomics revealed the protective effect of baicalin on alveolar bone inflammatory resorption in aging. *Phytomedicine* 2024;124:155233.
4. Pawar AR, Ramamurthy J, Girija ASS. Evaluation of antimicrobial susceptibility and resistance patterns of *Treponema denticola* isolated from periodontal disease: an in vitro study. *Cureus* 2024;16(6):e61497.
5. Ardila CM, Bedoya-García JA, Arrubla-Escobar DE. Antibiotic resistance in periodontitis patients: a systematic scoping review of randomized clinical trials. *Oral Dis* 2023;29(7):2501–11.
6. Bhuvrangan A, King R, Larvin H, Aggarwal VR. Antibiotic use and misuse in dentistry in India: a systematic review. *Antibiotics (Basel)* 2021;10(12):1459.
7. Arumuganainar D. A call for antibiotic stewardship initiatives for dentistry in India. *Lancet* 2024;404(10457):1014–5.
8. Dave M, Tattar R. Antimicrobial resistance genes in the oral microbiome. *Evid Based Dent* 2025;26:42–3. doi: 10.1038/s41432-025-01120-z.
9. Cao L, Yang H, Huang Z, et al. Direct prediction of antimicrobial resistance in *Pseudomonas aeruginosa* by metagenomic next-generation sequencing. *Front Microbiol* 2024;15:1413434.
10. Sauerborn E, Corredor NC, Reska T, et al. Detection of hidden antibiotic resistance through real-time genomics. *Nat Commun* 2024;15(1):5494.
11. El Naqa I, Ruan D, Valdes G, et al. Machine learning and modeling: data, validation, communication challenges. *Med Phys* 2018;45(10):e834–40.
12. Wu J, Ouyang J, Qin H, et al. PLM-ARG: antibiotic resistance gene identification using a pretrained protein language model. *Bioinformatics* 2023;39(11):btad690.
13. Li Y, Xu Z, Han W, et al. HMD-ARG: hierarchical multi-task deep learning for annotating antibiotic resistance genes. *Microbiome* 2021;9(1):40.
14. Yang Y, Jiang X, Chai B, et al. ARGs-OAP: online analysis pipeline for antibiotic resistance genes detection from metagenomic data using an integrated structured ARG-database. *Bioinformatics* 2016;32(15):2346–51.
15. Pinheiro ET, Karygianni L, Candeiro GTM, et al. Metatranscriptome and resistome of the endodontic microbiome. *J Endod* 2024;50(8):1059–72 e4.
16. Ardila CM, González-Arroyave D, Tobón S. Machine learning for predicting antimicrobial resistance in critical and high-priority pathogens: a systematic review considering antimicrobial susceptibility tests in real-world healthcare settings. *PLoS One* 2025;20(2):e0319460.
17. Consortium TU. UniProt: the universal protein knowledgebase in 2023. *Nucleic Acids Res* 2022;51(D1):D523–31.
18. Ardila CM, Yadalam PK, González-Arroyave D. Integrating whole genome sequencing and machine learning for predicting antimicrobial resistance in critical pathogens: a systematic review of antimicrobial susceptibility tests. *PeerJ* 2024;12:e18213.